Customer Churn Intelligence

Business Analytics Report

Generated on June 05, 2026

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1. Executive Summary

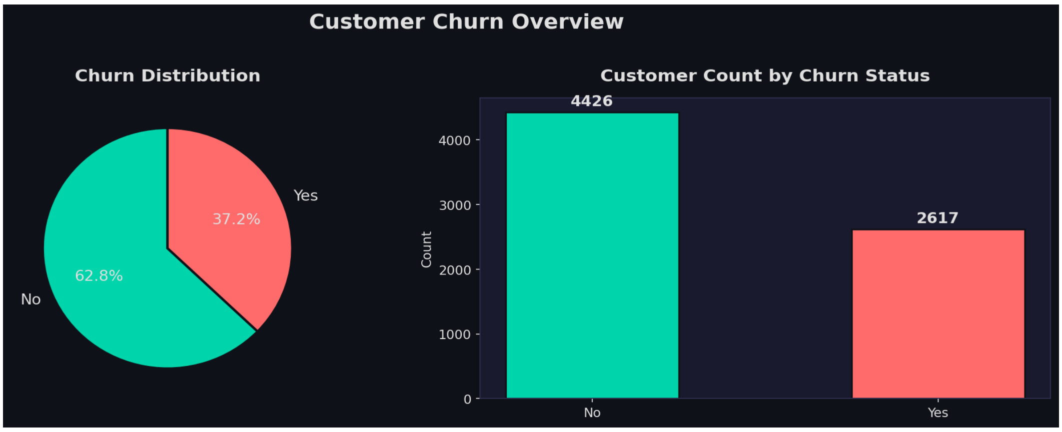
This report analyzes **7,043** customers to identify churn patterns, risk factors, and actionable retention strategies. The current churn rate stands at **37.2%**, with an estimated **$3,871,929** in revenue at risk.

Total Customers	Churn Rate	Revenue at Risk	Avg Tenure
7,043	37.2%	$3,871,929	26.1 months

2. Exploratory Data Analysis

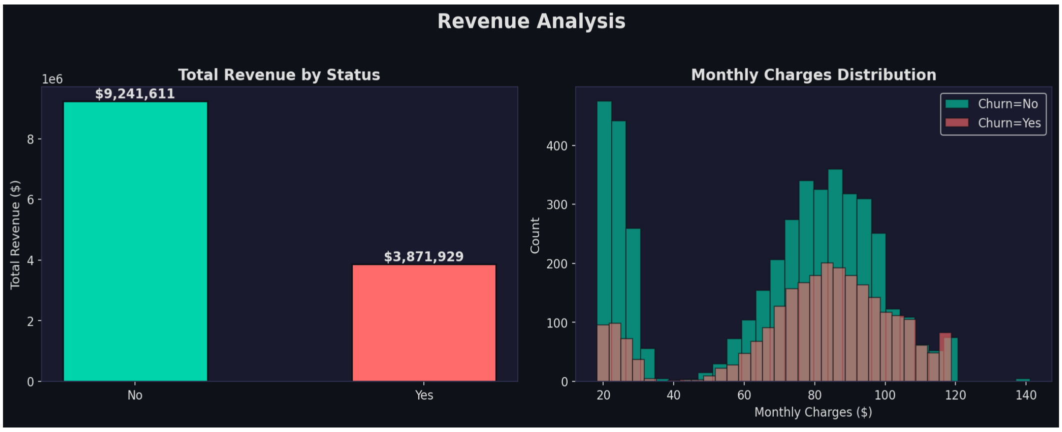
Churn Distribution

The overall churn rate shows a significant portion of customers leaving the service.



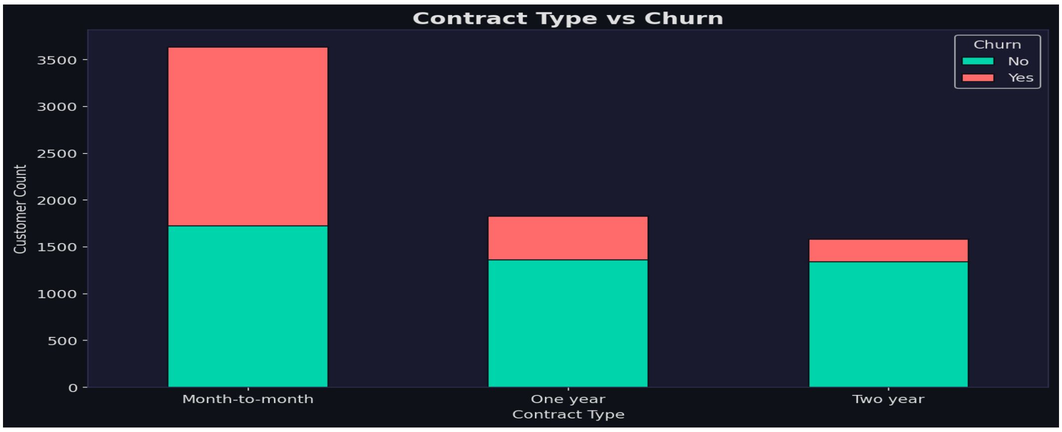
Revenue Analysis

Revenue loss from churned customers represents a substantial business impact.



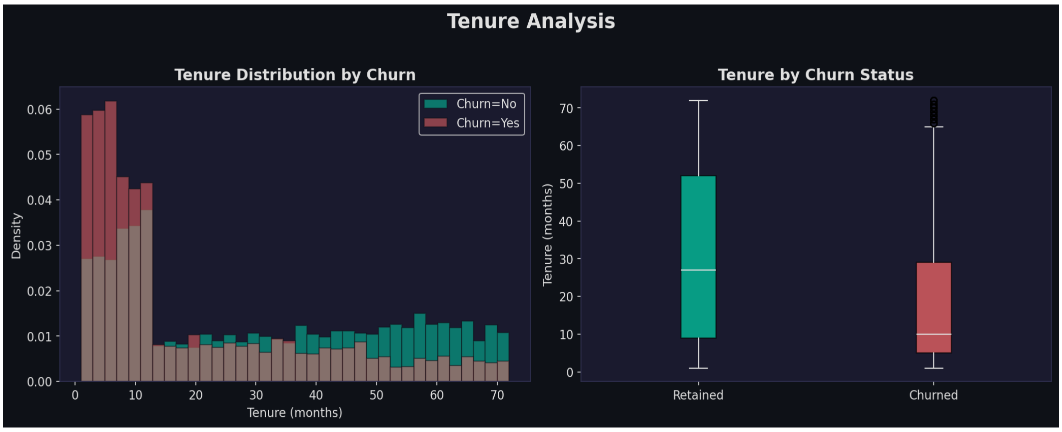
Contract Type Analysis

Month-to-month contracts show significantly higher churn rates compared to long-term contracts.



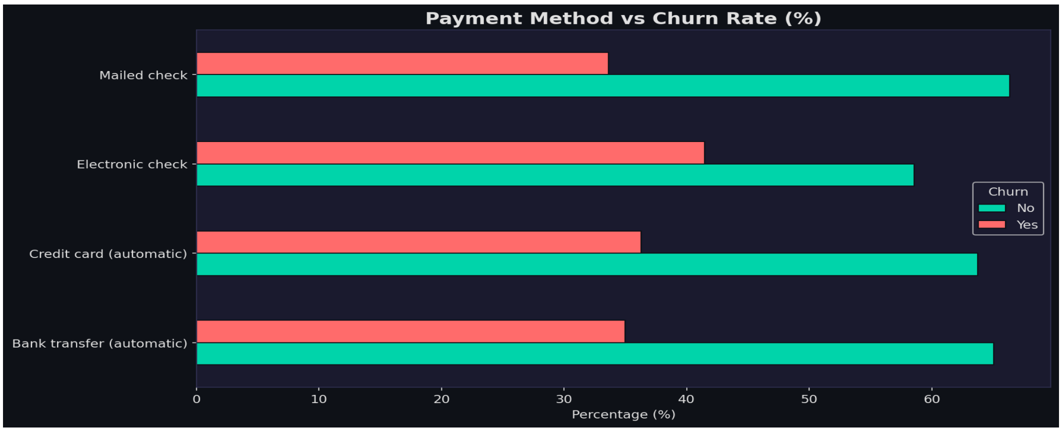
Tenure Analysis

Customers with shorter tenure are more likely to churn, indicating early engagement is critical.



Payment Method Impact

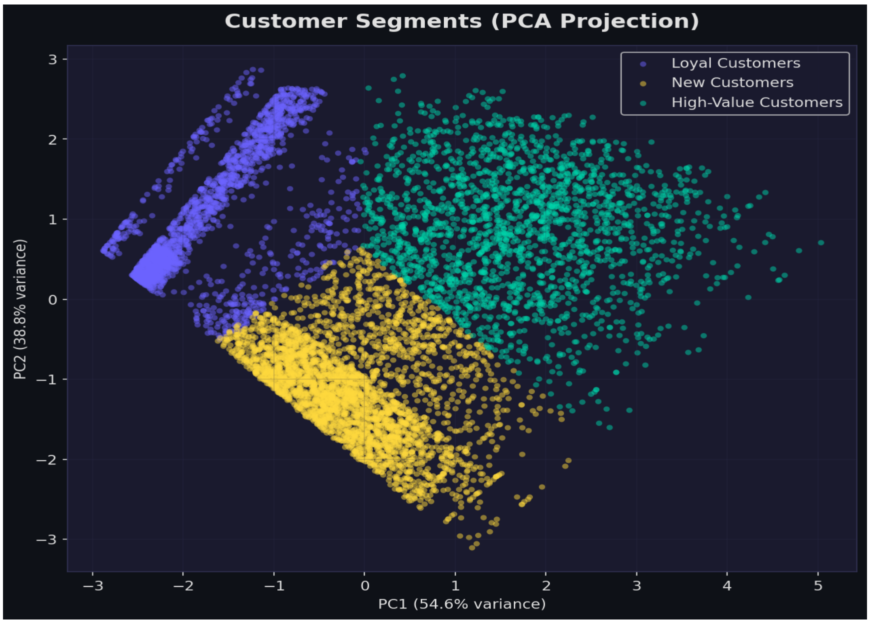
Electronic check users exhibit the highest churn rates, suggesting friction in the payment process.



3. Customer Segmentation

Customers were segmented using K-Means clustering based on tenure, monthly charges, total charges, and number of services. The analysis identified four distinct customer segments.

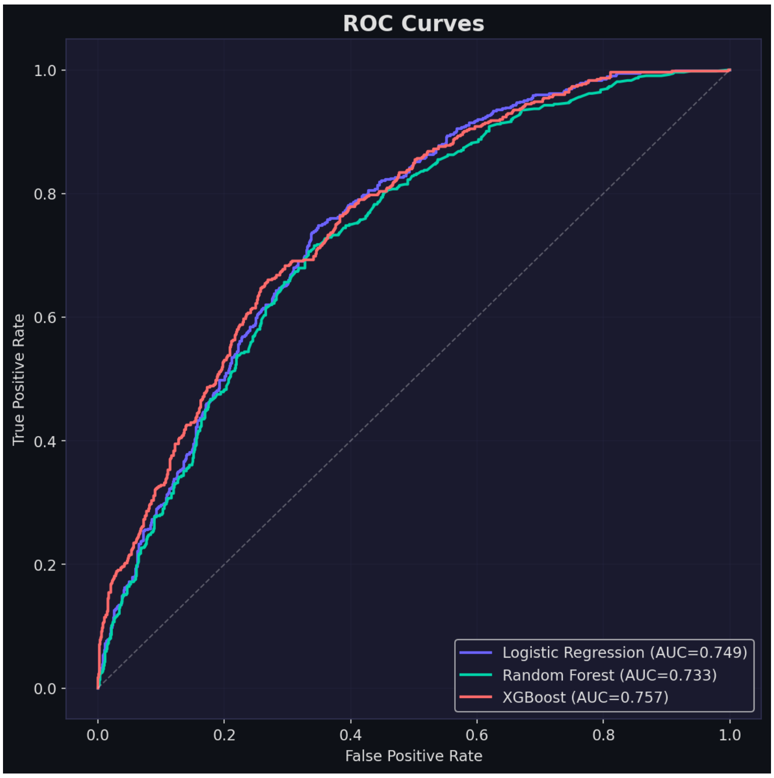
Segment	Count	Avg Tenure	Avg Monthly	Churn Rate
High-Value Customers	1989	51.9	$86.70	27.0%
Loyal Customers	1821	26.1	$29.39	24.0%
New Customers	3233	10.3	$86.89	51.0%



4. Prediction Model Performance

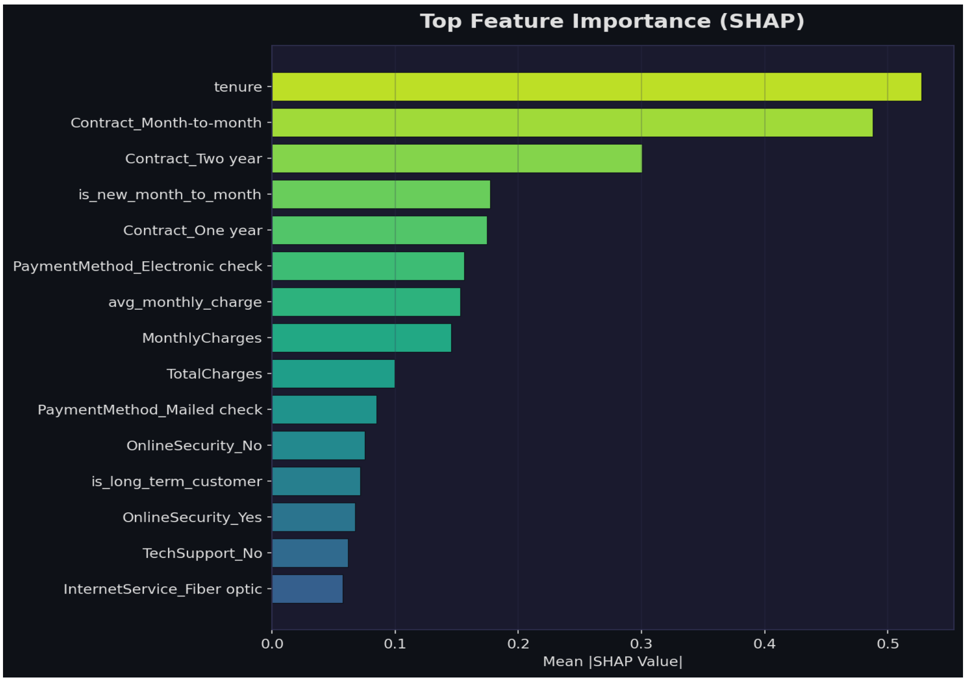
Three machine learning models were trained and evaluated: Logistic Regression, Random Forest, and XGBoost. The best model was selected based on F1 Score.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.6835	0.5609	0.6851	0.6168	0.7492
Random Forest	0.6849	0.5769	0.5725	0.5747	0.7334
XGBoost	0.7019	0.5945	0.6240	0.6089	0.7566



5. Key Insights & Churn Drivers

Feature Importance (SHAP)



Business Insights

1. **[Contract]** Customers on month-to-month contracts are 2.5x more likely to churn.
2. **[Tenure & Charges]** Customers with tenure < 12 months and monthly charges > $70 have a 56% churn rate.
3. **[Internet Service]** Fiber optic customers churn at 47% vs 36% for DSL — investigate service quality.
4. **[Demographics]** Senior citizens have a 40% churn rate vs 37% for non-seniors.
5. **[Payment]** Electronic check users churn at 41% vs 36% for auto-payment customers.
6. **[Model]** Top churn drivers: tenure, Contract_Month-to-month, Contract_Two year.

6. Recommendations

Introduce Loyalty Discounts [High Priority]

Offer loyalty discounts for customers with tenure under 12 months and monthly charges above $70. This targets the highest-risk segment.

Expected Impact: Could reduce churn by 15-20% in the at-risk segment.

Promote Long-Term Contracts [High Priority]

Incentivize month-to-month customers to switch to annual or biennial contracts with a 10-15% discount. Month-to-month contracts are the strongest predictor of churn.

Expected Impact: Potential to stabilize 40%+ of at-risk customers.

Improve Fiber Optic Experience [Medium Priority]

Fiber optic customers show higher churn rates despite paying more. Investigate service quality and consider value-add bundles.

Expected Impact: Address quality-price perception gap for premium customers.

Migrate from Electronic Check [Medium Priority]

Customers paying by electronic check churn significantly more. Offer auto-payment setup incentives (e.g., $5/month credit).

Expected Impact: Reduce friction-based churn in the payment process.

Bundle Security & Support Services [Medium Priority]

Customers without Online Security and Tech Support churn at higher rates. Create discounted bundles to improve engagement and reduce churn.

Expected Impact: Increase service stickiness and customer lifetime value.

Senior Citizen Retention Program [Low Priority]

Launch a dedicated senior-friendly support program with simplified billing and priority assistance.

Expected Impact: Improve retention in the senior citizen demographic.